

Periodic function or fundamental period

A function $f(x)$ is said to be periodic if and only if $f(x+p) = f(x)$ is true for some value of p and every value of x .

The smallest value of p for which this equation is true for every values of x this is called periodic of the function $f(x)$

Dirichlet's condition

(i) $f(x)$ is periodic, single valued and finite

(ii) $f(x)$ has a finite no. of discontinuity in any one period and no infinite discontinuity

(iii) $f(x)$ has at most a finite no. of maximum or minimum

Type I $\begin{cases} (0, 2\pi) \\ (-\pi, \pi) \end{cases}$

Change of Interval $\rightarrow$ Type II $\begin{cases} (0, 2l) \\ (-l, l) \end{cases}$

Half Range $\rightarrow$ Type III $\begin{cases} (0, \pi) \\ (0, l) \end{cases}$

Fourier series for $(0, 2\pi)$

The Fourier series for the fun $f(x)$

in the interval $(0, 2\pi)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin x$$

where $a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx$$

Deduction :

$(0, 2\pi)$

$x=0$ is an discontinuity pt

$\therefore$ Fourier series at $x=0$ converges to $\frac{f(0) + f(2\pi)}{2}$

$$(-\pi, \pi)$$

$$x=0$$

is continuity pt

$\therefore$ Fourier Series

at $x=0$ converges to $f(0)$.

(ii) $x=\pi$ is discontinuity pt

$\therefore$ F.S at $x=\pi$ converges to $\frac{f(-\pi) + f(\pi)}{2}$

problem:

$f(x) = (\pi - x)^2$ in $(0, 2\pi)$ deduce the

sum of series $1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots$

Soln

W.K.T

Fourier series for the fun $f(x)$ in the interval $(0, 2\pi)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx \rightarrow \textcircled{1}$$

find a_0 :

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} (\pi - x)^2 dx$$

$$= \frac{1}{\pi} \left[\frac{(\pi - x)^3}{-3} \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[\frac{-1}{3} (\pi - x)^3 \right]_0^{2\pi}$$

$$= \frac{-1}{3\pi} \left[(\pi - 2\pi)^3 - (\pi - 0)^3 \right]$$

$$= \frac{-1}{3\pi} \left[(-\pi)^3 - (\pi)^3 \right]$$

$$= \frac{-1}{3\pi} (-\pi^3 - \pi^3)$$

$$= \frac{-1}{3\pi} (-2\pi^3)$$

$$\boxed{a_0 = \frac{2\pi^2}{3}}$$

To find a_n

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx \, dx$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} (\pi - x)^2 \cos nx \, dx$$

By parts

$$u = (\pi - x)^2$$

$$dv = \cos nx$$

$$u' = -2(\pi - x)$$

$$v = \frac{\sin nx}{n}$$

$$u'' = -2(-1) = 2$$

$$v_1 = \frac{-\cos nx}{n^2}$$

$$v_2 = \frac{-\sin nx}{n^3}$$

W.K.T

$$\int u \, dv = uv - u'v_1 + u''v_2 \dots$$

$$a_n = \frac{1}{\pi} \left[(\pi - \alpha)^2 \frac{\sin n\alpha}{n} - 2(\pi - \alpha) \frac{\cos n\alpha}{n^2} - 2 \frac{\sin n\alpha}{n^3} \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left\{ \frac{1}{n} \left[(\pi - \alpha)^2 \sin n\alpha \right]_0^{2\pi} - \frac{2}{n^2} \left[(\pi - \alpha) \cos n\alpha \right]_0^{2\pi} \right\}$$

$$- \frac{2}{n^3} \left[\sin n\alpha \right]_0^{2\pi} \Bigg\}$$

$$\sin n\pi = 0$$

$$\sin n\pi = 0$$

$$\sin 0 = 0$$

$$\cos 2\pi = 1$$

$$\cos n\pi = (-1)^n$$

$$\cos 0 = 1$$

$$\cos(\pi - \pi) = -1$$

$$\cos(\pi - \pi) = -1$$

$$= \frac{1}{\pi} \left\{ \frac{1}{n} \left[(\pi - 2\pi)^2 \sin n\pi - (\pi - 0)^2 \sin n(0) \right] \right.$$

$$- \frac{2}{n^2} \left[(\pi - 2\pi) \cos n\pi - (\pi - 0) \cos n(0) \right]$$

$$- \frac{2}{n^3} \left[\sin n\pi - \sin n(0) \right] \Bigg\}$$

$$= \frac{1}{\pi} \left\{ \frac{-2}{n^2} \left[(\pi - 2\pi) - (\pi - 0) \right] \right.$$

$$= \frac{1}{\pi} \left[\frac{-2}{n^2} (-\pi - \pi) \right]$$

$$= \frac{1}{\pi} \left[\frac{-2}{n^2} (-2\pi) \right]$$

$$+ \frac{4}{n^2}$$

$$a_n = \frac{4}{n^2}$$

• find b_n

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx \, dx$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} (\pi - x)^2 \sin nx \, dx$$

$$u = (\pi - x)^2 \quad dv = \sin nx$$

$$u' = -2(\pi - x) \quad v = -\frac{\cos nx}{n}$$

$$u'' = -2(-1) = 2 \quad v_1 = -\frac{\sin nx}{n^2}$$

$$v_2 = \frac{\cos nx}{n^3}$$

$$\int u \, dv = uv - u'v_1 + u''v_2$$

$$b_n = \frac{1}{\pi} \left[-(\pi - x)^2 \frac{\cos nx}{n} - 2(\pi - x) \frac{\sin nx}{n^2} + \frac{2 \cos nx}{n^3} \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left\{ \left[-\frac{1}{n} [(\pi - x)^2 \cos nx] - \frac{2}{n^2} [(\pi - x) \sin nx] + \frac{2}{n^3} [\cos nx] \right]_0^{2\pi} \right\}$$

$$= \frac{1}{\pi} \left\{ -\frac{1}{n} [(\pi - 2\pi)^2 \cos 2\pi] - \frac{2}{n^2} [(\pi - 0)^2 \cos n(0)] \right\}$$

$$= \frac{1}{\pi} \left\{ -\frac{2}{n^2} [(\pi - 2\pi) \sin 2\pi] - \frac{2}{n^2} [(\pi - 0) \sin n(0)] \right\}$$

$$+ \frac{2}{n^3} \left[\underbrace{\cos n\pi}_1 - \underbrace{\cos n(0)}_1 \right] y$$

$$= \frac{1}{\pi} \left\{ \left[\frac{-1}{n} [(-\pi)^2 - (\pi)^2] + \frac{2}{n^3} (1-1) \right] y \right.$$

$$= \frac{1}{\pi} \left[\frac{-1}{n} \left(\frac{\pi^2 - \pi^2}{0} \right) \right]$$

$$= \frac{1}{\pi} \left[\frac{-1}{n} (0) \right]$$

$$= \frac{1}{\pi} \frac{0}{n}$$

$$b_n = 0$$

values in equ ①

Sub above

$$f(x) = \frac{1}{2} \left(\frac{\pi^2}{3} \right) + \sum_{n=1}^{\infty} \frac{4}{n^2} \cos nx + \sum_{n=1}^{\infty} (0) \sin nx$$

$$f(x) = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{\cos nx}{n^2}$$

Deduction:

$$\frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{\cos nx}{n^2} = f(x)$$

$$\frac{\pi^2}{3} + 4 \left[\frac{\cos x}{1^2} + \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} + \dots \right] = f(x)$$

$P_n(0, 2\lambda)$

Fourier series $x=0$ converges to $\frac{f(0) + f(2\lambda)}{2}$

$$\frac{\lambda^2}{3} + 4 \left[\frac{\cos 0}{1^2} + \frac{\cos 0}{2^2} + \frac{\cos 0}{3^2} + \dots \right] = \frac{f(0) + f(2\lambda)}{2}$$

$$f(x) = (\pi - x)^2$$

$$f(0) = (\pi - 0)^2 = \pi^2$$

$$f(2\lambda) = (\pi - 2\lambda)^2 = (-\lambda)^2 = \lambda^2$$

$$\frac{\lambda^2}{3} + 4 \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right] = \frac{\lambda^2 + \lambda^2}{2}$$

$$4 \left[\frac{1}{1^2} + \frac{1}{2^2} + \dots \right] = \frac{2\lambda^2}{2} - \frac{\lambda^2}{3}$$

$$4 \left[\frac{1}{1^2} + \frac{1}{2^2} + \dots \right] = \frac{3\lambda^2 - \lambda^2}{3}$$

$$4 \left[\frac{1}{1^2} + \frac{1}{2^2} + \dots \right] = \frac{2\lambda^2}{3}$$

$$\frac{1}{1^2} + \frac{1}{2^2} + \dots = \frac{2\lambda^2}{3} \times \frac{1}{4}$$

$$= \frac{\lambda^2}{6}$$

$f(x) = \left(\frac{\pi-x}{2}\right)^2$ in $(0, 2\pi)$. Deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{6}$
w.k.t

Fourier series for the fun $f(x)$ in the interval $(0, 2\pi)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx \rightarrow (1)$$

To find a_0 :

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$$
$$= \frac{1}{\pi} \int_0^{2\pi} \left(\frac{\pi-x}{2}\right)^2 dx$$

$$= \frac{1}{\pi} \cdot \frac{1}{4} \int_0^{2\pi} (\pi-x)^2 dx$$

$$= \frac{1}{4\pi} \left[\frac{(\pi-x)^3}{-3} \right]_0^{2\pi}$$

$$= \frac{-1}{12\pi} \left[(\pi-x)^3 \right]_0^{2\pi}$$

$$= \frac{-1}{12\pi} \left[(\pi-2\pi)^3 - (\pi-0)^3 \right]$$

$$= \frac{-1}{12\pi} \left[(-\pi)^3 - (\pi)^3 \right]$$

$$= \frac{-1}{12\pi} \left[-\pi^3 - \pi^3 \right]$$

$$= \frac{-1}{10\pi} (-2\pi^3)$$

$$a_0 = \pi^2/6$$

To find a_n

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx \, dx$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} \left(\frac{\pi-x}{2}\right)^2 \cos nx \, dx$$

$$= \frac{1}{4\pi} \int_0^{2\pi} (\pi-x)^2 \cos nx \, dx$$

$$u = (\pi-x)^2$$

$$dv = \cos nx$$

$$u' = -2(\pi-x)$$

$$v = \frac{\sin nx}{n}$$

$$u'' = -2(-1) = 2$$

$$v_1 = -\frac{\cos nx}{n^2}$$

$$v_2 = -\frac{\sin nx}{n^3}$$

$$a_n = \frac{1}{4\pi} \left[(\pi-x)^2 \frac{\sin nx}{n} - 2(\pi-x) \frac{\cos nx}{n^2} - 2 \frac{\sin nx}{n^3} \right]_0^{2\pi}$$

$$= \frac{1}{4\pi} \left\{ \frac{1}{n} [(\pi-x)^2 \sin nx]_0^{2\pi} - \frac{2}{n^2} [(\pi-x) \cos nx]_0^{2\pi} \right.$$

$$\left. - \frac{2}{n^3} [\sin nx]_0^{2\pi} \right\}$$

$$\begin{aligned}
 &= \frac{1}{4\pi} \int \left[\frac{1}{n} \left[\underbrace{(\pi-2\pi)^2}_{0} \underbrace{\sin n\pi}_{0} - \underbrace{(\pi-0)^2}_{0} \underbrace{\sin n(0)}_{0} \right] \right. \\
 &\quad \left. - \frac{2}{n^2} \left[(\pi-2\pi) \underbrace{\cos n\pi}_1 - (\pi-0) \underbrace{\cos n(0)}_1 \right] \right. \\
 &\quad \left. - \frac{2}{n^3} \left[\underbrace{\sin n\pi}_0 - \underbrace{\sin n(0)}_0 \right] \right] d\pi
 \end{aligned}$$

$$= \frac{1}{4\pi} \int \left[\frac{-2}{n^2} (-\pi - \pi) \right] d\pi$$

$$= \frac{1}{4\pi} \left[\frac{-2}{n^2} (-2\pi) \right]$$

$$a_n = \frac{1}{n^2}$$

to find b_n

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx \, dx$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} \left(\frac{\pi-x}{2} \right)^2 \sin nx \, dx$$

$$= \frac{1}{4\pi} \int_0^{2\pi} (\pi-x)^2 \sin nx \, dx$$

$$u = (\pi-x)^2 \quad \begin{array}{l} dv = \sin nx \\ v = \frac{-\cos nx}{n} \end{array}$$

$$u' = -2(\pi-x) \quad \begin{array}{l} v_1 = \frac{-\sin nx}{n^2} \end{array}$$

$$u'' = -2(-1) = 2 \quad \begin{array}{l} v_2 = \frac{\cos nx}{n^3} \end{array}$$

$$= \frac{1}{4\pi} \left\{ -\frac{(\pi-x)^2 \cos nx}{n} - 2(\pi-x) \frac{\sin nx}{n^2} + 2 \frac{\cos nx}{n^3} \right\}$$

$$= \frac{1}{4\pi} \left\{ -\frac{1}{n} \left[(\pi-x)^2 \cos nx \right]_0^{2\pi} - \frac{2}{n^2} \left[(\pi-x) \sin nx \right]_0^{2\pi} + \frac{2}{n^3} \left[\cos nx \right]_0^{2\pi} \right\}$$

$$= \frac{1}{4\pi} \left\{ -\frac{1}{n} \left[(\pi-2\pi)^2 \cos n2\pi - (\pi-0)^2 \cos n(0) \right] - \frac{2}{n^2} \left[(\pi-2\pi) \sin n2\pi - (\pi-0) \sin n(0) \right] + \frac{2}{n^3} \left[\cos n2\pi - \cos n(0) \right] \right\}$$

$$= \frac{1}{4\pi} \left\{ -\frac{1}{n} \left[(-\pi)^2 - (\pi)^2 \right] + \frac{2}{n^3} (1-1) \right\}$$

$$= \frac{1}{4\pi} \left[-\frac{1}{n} (+\pi^2 - \pi^2) \right]$$

$$= \frac{1}{4\pi} \left[-\frac{1}{n} (0) \right]$$

$$b_n = 0$$

Sub above value in equ (1)

$$f(x) = \frac{1}{2} \left(\frac{\pi^2}{6} \right) + \sum_{n=1}^{\infty} \frac{1}{n^2} \cos nx + \sum_{n=1}^{\infty} (0) \sin nx$$

$$f(x) = \frac{1}{2} \left(\frac{\pi^2}{6} \right) + \sum_{n=1}^{\infty} \frac{\cos nx}{n^2}$$

Reduction:

$$\frac{\pi^2}{12} + \sum_{n=1}^{\infty} \frac{\cos nx}{n^2} = f(x)$$

$$\frac{\pi^2}{12} + \left[\frac{\cos x}{1^2} + \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} + \dots \right] = f(x)$$

$f_n(0, 2\pi)$

Fourier series $x=0$ converges to $\frac{f(0) + f(2\pi)}{2}$

$$\frac{\pi^2}{12} + \left[\frac{\cos 0}{1^2} + \frac{\cos 0}{2^2} + \frac{\cos 0}{3^2} + \dots \right] = \frac{f(0) + f(2\pi)}{2}$$

$$f(x) = \left(\frac{\pi - x}{2} \right)^2$$

$$f(0) = \left(\frac{\pi - 0}{2} \right)^2 = \frac{\pi^2}{4}$$

$$f(2\pi) = \left(\frac{\pi - 2\pi}{2} \right)^2 = \left(\frac{-\pi}{2} \right)^2 = \frac{\pi^2}{4}$$

$$\frac{\pi^2}{12} + \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right] = \frac{\pi^2/4 + \pi^2/4}{2}$$

$$\frac{\pi^2}{12} + \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right] = 2 \frac{\pi^2}{8}$$

$$\left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right] = \frac{2\pi^2}{8} - \frac{\pi^2}{12}$$

$$\begin{aligned}
 \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots &= \frac{\pi^2}{4} - \frac{\pi^2}{12} \\
 &= \frac{3\pi^2 - \pi^2}{12} \\
 &= \frac{2\pi^2}{12} \\
 &= \pi^2/6
 \end{aligned}$$

2. $f(x) = x(2\pi - x)$ in $(0, 2\pi)$ Deduce that

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \pi^2/6$$

Solu.

w.k.T

Fourier series for the fun $f(x)$ in the interval $(0, 2\pi)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx \rightarrow \textcircled{1}$$

To find a_0

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$$

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} x(2\pi - x) dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} (2\pi x - x^2) dx$$

$$= \frac{1}{\pi} \left(2\pi \frac{x^2}{2} - \frac{x^3}{3} \right)_0^{2\pi}$$

$$= \frac{1}{\pi} \left[\left(\pi(2\pi)^2 - \frac{(2\pi)^3}{3} \right) - (0) \right]$$

$$= \frac{1}{\pi} \left(4\pi^3 - \frac{8\pi^3}{3} \right)$$

$$= \frac{1}{\pi} \left(\frac{12\pi^3 - 8\pi^3}{3} \right)$$

$$= \frac{1}{\pi} \frac{4\pi^3}{3}$$

$$a_0 = \frac{4\pi^2}{3}$$

To find a_n :

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx \, dx$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} x(2\pi - x) \cos nx \, dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} (2\pi x - x^2) \cos nx \, dx$$

$$u = (2\pi x - x^2)$$

$$dv = \cos nx$$

$$u' = (2\pi - 2x)$$

$$v = \frac{\sin nx}{n}$$

$$u'' = -2$$

$$v_1 = -\frac{\cos nx}{n^2}$$

$$v_2 = -\frac{\sin nx}{n^3}$$

$$a_n = \frac{1}{\pi} \left[(2\pi x - x^2) \frac{\sin nx}{n} + (2\pi - 2x) \frac{\cos nx}{n^2} + 2 \frac{\sin nx}{n^3} \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left\{ \frac{1}{n} \left[(2\pi x - x^2) \sin nx \right]_0^{2\pi} + \frac{1}{n^2} \left[(2\pi - 2x) \cos nx \right]_0^{2\pi} + \frac{2}{n^3} \left[\sin nx \right]_0^{2\pi} \right\}$$

$$= \frac{1}{\pi} \left\{ \frac{1}{n} \left[(2\pi(2\pi) - (2\pi)^2) \sin n(2\pi) - (2\pi(0) - 0) \sin n(0) \right] + \frac{1}{n^2} \left[(2\pi - 2(2\pi)) \cos n(2\pi) - (2\pi - 2(0)) \cos n(0) \right] + \frac{2}{n^3} \left[\sin n(2\pi) - \sin n(0) \right] \right\}$$

$$= \frac{1}{\pi} \left\{ \frac{1}{n^2} \left[(2\pi - 4\pi) (1) - (2\pi) (1) \right] \right\}$$

$$= \frac{1}{\pi} \left[\frac{1}{n^2} (-2\pi - 2\pi) \right]$$

$$= \frac{1}{\pi} \left[\frac{1}{n^2} (-4\pi) \right]$$

$$a_n = \frac{-4}{n^2}$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx \, dx$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} (2\pi x - x^2) \sin nx \, dx$$

$$u = (2\pi x - x^2)$$

$$dv = \sin nx$$

$$u' = (2\pi - 2x)$$

$$v = \frac{-\cos nx}{n}$$

$$u'' = -2$$

$$v_1 = -\frac{\sin nx}{n^2}$$

$$v_2 = \frac{\cos nx}{n^3}$$

$$b_n = \frac{1}{\pi} \left\{ - (2\pi x - x^2) \frac{\cos nx}{n} + (2\pi - 2x) \frac{\sin nx}{n^2} - 2 \frac{\cos nx}{n^3} \right\}_0^{2\pi}$$

$$= \frac{1}{\pi} \left\{ -\frac{1}{n} \left[(2\pi x - x^2) \cos nx \right]_0^{2\pi} + \frac{1}{n^2} \left[(2\pi - 2x) \sin nx \right]_0^{2\pi} - \frac{2}{n^3} \left[\cos nx \right]_0^{2\pi} \right\}$$

$$= \frac{1}{\pi} \left\{ -\frac{1}{n} \left[(2\pi(2\pi) - (2\pi)^2) \cos n(2\pi) - (2\pi(0) - 0) \cos n(0) \right] \right.$$

$$+ \frac{1}{n^2} \left[(2\pi - 2(2\pi)) \sin n(2\pi) - (2\pi - 2(0)) \sin n(0) \right]$$

$$- \frac{2}{n^3} \left[\cos n(2\pi) - \cos n(0) \right]$$

$$= \frac{1}{\pi} \left\{ \frac{1}{n} \left[(4\pi^2 - \cancel{4\pi^2}) - 2\pi \right] - \frac{2}{n^3} (1) \right\}$$

$$b_n = 0$$

Sub above values in equ (1)

$$f(x) = \frac{1}{2} \left(\frac{4\pi^2}{3} \right) + \sum_{n=1}^{\infty} \left(\frac{-4}{n^2} \right) \cos nx + \sum_{n=1}^{\infty} (0) \sin nx$$

$$f(x) = \frac{2\pi^2}{3} - 4 \sum_{n=1}^{\infty} \frac{\cos nx}{n^2}$$

Deduction:

$$\frac{2\pi^2}{3} - 4 \sum_{n=1}^{\infty} \frac{\cos nx}{n^2} = f(x)$$

$$\frac{2\pi^2}{3} - 4 \left[\frac{\cos x}{1^2} + \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} + \dots \right] = f(x)$$

In $(0, 2\pi)$

Fourier Series $x=0$ converges to $\frac{f(0) - f(2\pi)}{2}$

$$f(x) = 2\pi x - x^2$$

$$f(0) = 2\pi(0) - 0 = 0$$

$$f(2\pi) = 2\pi(2\pi) - (2\pi)^2 = 4\pi^2 - 4\pi^2 = 0$$

$$\frac{2\pi^2}{3} - 4 \left[\frac{\cos 0}{1^2} + \frac{\cos 0}{2^2} + \frac{\cos 0}{3^2} + \dots \right] = \frac{f(0) + f(2\pi)}{2}$$

$$\frac{2\pi^2}{3} - 4 \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right] = \frac{0+0}{2}$$

$$\frac{2\pi^2}{3} - 4 \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right] = 0$$

$$\frac{2\pi^2}{3} = 4 \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right]$$

$$\frac{2\pi^2}{3} \times \frac{1}{4} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots$$

$$\frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots$$

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{6}$$

30/9/15 4. $f(x) = x^2$ in $(-\pi, \pi)$ hence deduce

that $1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots$

$$1 - \frac{1}{2^2} + \frac{1}{3^2} + \dots$$

G.T.

$$f(x) = x^2$$

$$f(-x) = (-x)^2$$

$$= x^2$$

Even fun

$$f(x) = f(-x)$$

odd fun

$$f(x) = -f(-x)$$

$$f(x) = +f(-x)$$

$f(x)$ is even fun.

W.K.T

Fourier series for $(-\pi, \pi)$ is

Here $f(x)$ is an even fun then

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx \rightarrow \textcircled{A}$$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$$

To find a_0 :

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx$$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} x^2 dx$$

$$= \frac{2}{\pi} \left(\frac{x^3}{3} \right)_0^{\pi}$$

$$= \frac{2}{3\pi} (x^3)_0^{\pi}$$

$$= \frac{2}{3\pi} (\pi^3 - 0)$$

$$a_0 = \frac{2\pi^2}{3}$$

$$\int_{-\infty}^{\infty} = 2 \int_0^{\infty}$$

$$\int_{-\pi}^{\pi} = 2 \int_0^{\pi}$$

find a_n :

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} x^2 \cos nx dx$$

$$u = x^2$$

$$dv = \cos nx$$

$$u' = 2x$$

$$v_2 = \frac{\sin nx}{n}$$

$$u'' = 2$$

$$v_1 = -\frac{\cos nx}{n^2}$$

$$v_2 = -\frac{\sin nx}{n^3}$$

$$\int u dv = uv - u'v_1 + u''v_2$$

$$a_n = \frac{2}{\pi} \left[x^2 \frac{\sin nx}{n} + \frac{2x \cos nx}{n^2} - \frac{2 \sin nx}{n^3} \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[\frac{1}{n} (x^2 \sin nx)_0^{\pi} + \frac{2}{n^2} (x \cos nx)_0^{\pi} - \frac{2}{n^3} (\sin nx)_0^{\pi} \right]$$

$$= \frac{2}{\pi} \left[\frac{1}{n} (\lambda^2 \sin n\lambda - 0) + \frac{2}{n^3} (\lambda \cos n\lambda - 0) \right. \\ \left. - \frac{2}{n^3} (\sin n\lambda - \sin 0) \right]$$

$$= \frac{2}{\pi} \left[\frac{2}{n^2} \lambda (-1)^n \right] \Rightarrow \frac{2}{\pi} \left[\frac{2}{n^2} \lambda (-1)^n \right]$$

$$a_n = 2 \left(\frac{2}{n^2} \right) (-1)^n$$

$$(-1)^n = \cos n\pi$$

$$a_n = \frac{4}{n^2} (-1)^n$$

Deduction:

Sub a_0 and a_n in equ (1)

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cosh nx$$

$$= \frac{1}{2} \left(\frac{2\pi^2}{3} \right) + \sum_{n=1}^{\infty} \frac{4}{n^2} (-1)^n \cosh nx$$

$$f(x) = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} (-1)^n \frac{\cosh nx}{n^2}$$

Deduction 1:

$$\frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} (-1)^n \frac{\cosh nx}{n^2} = f(x)$$

$$(-\pi, \pi)$$

1. F.S at $x=0$ eqs to $f(0)$ ✓

2. F.S at $x=\pi$ eqs to $\frac{f(-\pi) + f(\pi)}{2}$ ✓

$$g_n(-\pi, \pi)$$

Fourier series $x=0$

converges to $f(0)$

$$f(x) = x^2$$

$$f(0) = 0$$

$$\frac{\pi^2}{3} + 4 \left[\underbrace{-\frac{\cos \pi}{1^2}}_{n=1} + \underbrace{\frac{\cos 2\pi}{2^2}}_{n=2} - \underbrace{\frac{\cos 3\pi}{3^2}}_{n=3} \dots \right] = f(x)$$

$$\frac{\pi^2}{3} + 4 \left[-\frac{1}{1^2} + \frac{1}{2^2} - \frac{1}{3^2} \dots \right] = 0$$

$$4 \left[-\frac{1}{1^2} + \frac{1}{2^2} - \frac{1}{3^2} \dots \right] = -\frac{\pi^2}{3}$$

$$-\frac{1}{1^2} + \frac{1}{2^2} - \frac{1}{3^2} = -\frac{\pi^2}{3} \times \frac{1}{4}$$

$$\left[-\frac{1}{1^2} + \frac{1}{2^2} - \frac{1}{3^2} \dots \right] = -\frac{\pi^2}{12}$$

$$+ \left[\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} \dots \right] = +\frac{\pi^2}{12}$$

$$\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{12}$$

Deduction: 2

$$\frac{\pi^2}{3} + 4 \left[-\frac{\cos x}{1^2} + \frac{\cos 2x}{2^2} - \frac{\cos 3x}{3^2} \dots \right] = f(x)$$

$$\Sigma_n(-\pi, \pi)$$

f.s at $x=\pi$ converges at $\frac{f(-\pi) + f(\pi)}{2}$

$$f(x) = x^2$$

$$f(-\pi) = (-\pi)^2 = \pi^2$$

$$f(\pi) = \pi^2 = \pi^2$$

$$\frac{\pi^2}{3} + 4 \left[-\frac{\cos \pi}{1^2} + \frac{\cos 2\pi}{2^2} - \frac{\cos 3\pi}{3^2} \dots \right] = \frac{f(-\pi) + f(\pi)}{2}$$

$$\frac{\pi^2}{3} + 4 \left[-\frac{(-1)}{1^2} + \frac{1}{2^2} - \frac{(-1)}{3^2} \dots \right] = \frac{\pi^2 + \pi^2}{2}$$

$$\frac{\pi^2}{3} + 4 \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right] = \frac{2\pi^2}{2}$$

$$4 \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right] = \pi^2 - \frac{\pi^2}{3}$$

$$4 \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right] = \frac{3\pi^2 - \pi^2}{3} = \frac{2\pi^2}{3}$$

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{2\pi^2}{3} \times \frac{1}{4}$$

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{6}$$

$f(x) = |x|$ in $-\pi < x < \pi$ hence deduce
that $\frac{1}{1^2} + \frac{1}{3^2} + \dots = \pi^2/8$

G.T

$$f(x) = |x|$$

$$f(-x) = |-x| = |x|$$

$$f(x) = f(-x)$$

$\therefore f(x)$ is even function

w.k.T

Here $f(x)$ is an even fn then
Fourier series for $(-\pi, \pi)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx \rightarrow (1)$$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$$

To find a_0 :

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} |x| dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x dx$$

(by def of modulus)

$$|x| = \begin{cases} x, & x > 0 \\ -x, & x < 0 \end{cases}$$

$$= \frac{2}{\pi} \left(\frac{x^2}{2} \right)_0^{\pi}$$

$$= \frac{2}{\pi} \cdot \frac{1}{2} (x^2)_0^{\pi}$$

$$= \frac{1}{\pi} (\pi^2 - 0)$$

$$\boxed{a_0 = \pi}$$

To find a_n

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx$$

$$= \frac{2}{\pi} \int_0^{\pi} |x| \cos nx \, dx$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} x \cos nx \, dx$$

$$u = x$$

$$dv = \cos nx$$

$$u' = 1$$

$$V = \frac{\sin nx}{n}$$

$$V_1 = -\frac{\cos nx}{n^2}$$

$$a_n = \frac{2}{\pi} \left[\frac{x \sin nx}{n} + \frac{\cos nx}{n^2} \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[\frac{1}{n} (x \sin nx)_0^{\pi} + \frac{1}{n^2} (\cos nx)_0^{\pi} \right]$$

Even
 $n=2$

$$a_n = \frac{2}{\pi n^2} [(-1)^n - 1]$$

$$= \frac{2}{\pi n^2} [0]$$

$$a_n = 0$$

$$n=3$$

$$a_n = \frac{2}{\pi n^2} [(-1)^n - 1]$$

$$= \frac{2}{\pi n^2} [(-1)^3 - 1]$$

$$= \frac{2}{\pi n^2} [-2]$$

$$= \frac{2}{\pi} \left[\frac{1}{n} (\cancel{x \sin nx} - \cancel{0 \sin n0}) + \frac{1}{n^2} (\cancel{\cos nx} - \cancel{\cos 0}) \right]$$

$$= \frac{2}{\pi} \left[\frac{1}{n^2} ((-1)^n - 1) \right]$$

$$a_n = \frac{2}{\pi n^2} [(-1)^n - 1] \rightarrow a_n = \begin{cases} 0, & n \text{ is even} \\ -\frac{4}{\pi n^2}, & n \text{ is odd.} \end{cases}$$

Sub a_0 and a_n in equ (1)

$$f(x) = \frac{1}{2} (a_0) + \sum_{n=1}^{\infty} a_n \cos nx$$

$$= \frac{\pi}{2} - \sum_{n=1,3,5} \frac{4}{\pi n^2} \cos nx$$

$$f(x) = \frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1,3,5} \frac{\cos nx}{n^2}$$

Deduction:

$$\frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1,3,5} \frac{\cos nx}{n^2} = f(x)$$

F.S at $x=0$ eqs to $f(0)$

$$\frac{\pi}{2} - \frac{4}{\pi} \left[\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \dots \right] = f(x)$$

$$\frac{\pi}{2} - \frac{4}{\pi} \left[\frac{\cos 0}{1^2} + \frac{\cos 0}{3^2} + \frac{\cos 0}{5^2} + \dots \right] = f(0)$$

$$f(x) = f(x)$$

$$f(0) = f(0) = 0$$

$$-\frac{4}{\pi} \left[\frac{1}{1^2} + \frac{1}{3^2} + \dots \right] = 0 - \pi/2$$

$$\frac{1}{1^2} + \frac{1}{3^2} + \dots = -\frac{\pi}{2} \times -\frac{\pi}{4}$$

$$\frac{1}{1^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{8}$$

Type III (change of interval) $\rightarrow \downarrow$

$$n\pi x \rightarrow \frac{n\pi x}{l}$$

$(0, 2l)$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}$$

$$a_0 = \frac{1}{l} \int_0^{2l} f(x) dx$$

$$a_n = \frac{1}{l} \int_0^{2l} f(x) \cos \frac{n\pi x}{l} dx$$

$$b_n = \frac{1}{l} \int_0^{2l} f(x) \sin \frac{n\pi x}{l} dx$$

$$l = 1, 2, 3, \dots$$

Problem:

$$f(x) = \begin{cases} \pi x & , 0 < x < 1 \\ \pi(2-x) & , 1 < x < 2 \end{cases}$$

$$2l = 2$$

$$l = 1$$

W.K.T

Fourier Series for $(0, 2l)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l} + \sum_{n=1}^{\infty} b_n \frac{\sin n\pi x}{l}$$

Here $l = 1$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos n\pi x + \sum_{n=1}^{\infty} b_n \frac{\sin n\pi x}{l} \rightarrow \textcircled{1}$$

find a_0 :

$$a_0 = \frac{1}{l} \int_0^{2l} f(x) dx$$

Here $l = 1$

$$a_0 = \frac{1}{1} \int_0^2 f(x) dx$$

$$= \left[\int_0^1 (\pi x) dx + \int_1^2 \pi(2-x) dx \right]$$

$$= \left[\pi \left(\frac{x^2}{2} \right)_0^1 + \left[2\pi x - \pi \left(\frac{x^2}{2} \right)_1^2 \right] \right]$$

$$= \left[\pi \left(\frac{1}{2} - 0 \right) + 2\pi(2-1) - \pi \left(\frac{4}{2} - \frac{1}{2} \right) \right]$$

$$= \frac{\bar{\lambda}}{2} + \left[2\lambda(1) - \lambda\left(\frac{3}{2}\right) \right]$$

$$= \frac{\bar{\lambda}}{2} + \lambda\left(2 - \frac{3}{2}\right)$$

$$= \frac{\bar{\lambda}}{2} + \lambda\left(\frac{4-3}{2}\right)$$

$$= \frac{\bar{\lambda}}{2} + \frac{\bar{\lambda}}{2} = \frac{2\bar{\lambda}}{2}$$

$$\boxed{a_0 = \bar{\lambda}}$$

To find a_n

$$a_n = \frac{1}{l} \int_0^{2l} f(x) \frac{\cosh n\pi x}{l} dx$$

Here $l=1$

$$a_n = \frac{1}{1} \int_0^2 f(x) \cosh n\pi x dx$$

$$= \left[\int_0^1 (\pi x) \cosh n\pi x dx + \int_1^2 \lambda(2-x) \cosh n\pi x dx \right]$$

$$u = \pi x$$

$$u' = \pi$$

$$u'' = 0$$

$$dv = \cosh n\pi x$$

$$v = \frac{\sinh n\pi x}{n\pi}$$

$$v_1 = \frac{-\cosh n\pi x}{n^2 \pi^2}$$

$$u = 2 - \pi(2-x)$$

$$u' = -\pi$$

$$u'' = 0$$

$$\begin{aligned}
& \left[\left(\frac{\bar{\lambda} x \sinh \bar{\lambda} x}{n \bar{\lambda}} + \frac{\bar{\lambda} \cosh \bar{\lambda} x}{n^2 \bar{\lambda}^2} \right)' + \left(\frac{\bar{\lambda} (2-x) \sinh \bar{\lambda} x}{n \bar{\lambda}} - \frac{\bar{\lambda} \cosh \bar{\lambda} x}{n^2 \bar{\lambda}^2} \right)' \right]_0^1 \\
& + \left[\left(\frac{\bar{\lambda}}{n \bar{\lambda}} (2 \sinh \bar{\lambda} x) \right)' + \frac{\bar{\lambda}}{n^2 \bar{\lambda}^2} (\cosh \bar{\lambda} x)' \right]_0^1 + \left[\left(\frac{\bar{\lambda}}{n \bar{\lambda}} ((2-x) \sinh \bar{\lambda} x)^2 - \frac{\bar{\lambda}}{n^2 \bar{\lambda}^2} (\cosh \bar{\lambda} x)^2 \right)' \right]_0^1 \\
& = \left[\frac{\bar{\lambda}}{n \bar{\lambda}} (1 \sinh \bar{\lambda} - 0) + \frac{\bar{\lambda}}{n^2 \bar{\lambda}^2} (\cosh \bar{\lambda} - \cosh \bar{\lambda}(0)) \right] \\
& + \left[\frac{\bar{\lambda}}{n \bar{\lambda}} ((2-2) \sinh \bar{\lambda}(2) - (2-1) \sinh \bar{\lambda}(1)) \right. \\
& \quad \left. - \left(\frac{\bar{\lambda}}{n^2 \bar{\lambda}^2} (\cosh \bar{\lambda}(2) - \cosh \bar{\lambda}(1)) \right) \right] \\
& = \left[\frac{\bar{\lambda}}{n^2 \bar{\lambda}^2} ((-1)^n - 1) - \left\{ \frac{\bar{\lambda}}{n^2 \bar{\lambda}^2} ((-1)^n - (-1)^n) \right\} \right] \\
& = \frac{\bar{\lambda}}{n^2 \bar{\lambda}^2} [(-1)^n - 1 - 1 + (-1)^n] \\
& = \frac{\bar{\lambda}}{n^2 \bar{\lambda}^2} [2(-1)^n - 2] \\
& = \frac{2\bar{\lambda}}{n^2 \bar{\lambda}^2} [(-1)^n - 1]
\end{aligned}$$

$$a_n = \begin{cases} 0, & n \text{ is even} \\ \frac{-4\lambda}{n^2\lambda^2}, & n \text{ is odd} \end{cases}$$

$$a_n = \frac{-4\lambda}{n^2\lambda^2} = \frac{-4}{n^2\lambda}$$

To find b_n :

$$b_n = \frac{1}{l} \int_0^{2l} f(x) \sin \frac{n\pi x}{l} dx$$

Here $l=1$

$$b_n = \frac{1}{1} \int_0^2 f(x) \sin n\pi x dx$$

$$b_n = \left[\int_0^1 (\pi x) \sin n\pi x dx + \int_1^2 \pi(2-x) \sin n\pi x dx \right]$$

$$u = \pi x$$

$$dv = \sin n\pi x$$

$$u = \pi(2-x)$$

$$u' = \pi$$

$$v = \frac{-\cos n\pi x}{n\pi}$$

$$u' = \pi(-1)$$

$$v'' = 0$$

$$v_1 = \frac{-\sin n\pi x}{n^2\pi^2}$$

$$= -\pi$$

$$= \left[-\left(\pi x \frac{\cos n\pi x}{n\pi} \right)' + \left(\pi \frac{\sin n\pi x}{n^2\pi^2} \right)' \right]_0^1$$

$$- \left(\pi(2-x) \frac{\cos n\pi x}{n\pi} \right)' + \left(\pi \frac{\sin n\pi x}{n^2\pi^2} \right)' \Bigg]_1^2$$

$$\begin{aligned}
 & \left[-\frac{\pi}{n\lambda} \left(\frac{(1) \cos n\lambda}{(-1)^2} - 0 \right) + \frac{\pi}{h^2 \lambda^2} \left(\sin n\lambda - \sin 0 \right) \right. \\
 & - \frac{\pi}{n\lambda} \left(\frac{(2-2) \cos 2n\lambda}{0} - (2-1) \frac{\cos n\lambda}{(-1)^2} \right) \\
 & \left. - \frac{\pi}{h^2 \lambda^2} \left(\sin 2n\lambda - \sin n\lambda \right) \right]
 \end{aligned}$$

$$= -\frac{\pi}{n\lambda} \left((-1)^n \right) - \frac{\pi}{n\lambda} \left(-(-1)^n \right)$$

$$= \frac{\pi}{n\lambda} \left[-(-1)^n + (-1)^n \right]$$

$$b_n = 0$$

Sub above values in equ (1)

$$f(x) = \frac{\pi}{2} - \frac{4}{\lambda} \sum_{n=1,3,5,\dots} \frac{\cosh n\pi x}{n^2} + \sum_{n=1}^{\infty} (0) \sin n\pi x$$

$$f(x) = \frac{\pi}{2} - \frac{4}{\lambda} \sum_{n=1,3,5,\dots} \frac{\cosh n\pi x}{n^2} //$$

2. $f(x) = x$ in $(-l, l)$ $\rightarrow$ $\boxed{n\pi = \frac{n\pi x}{l}}$

$$f(x) = x$$

$$f(-x) = -x$$

$$f(x) = -f(-x)$$

$\therefore f(x)$ is odd fn.

W.K.T

Fourier series in $(-l, l)$ is

For odd fn

$$f(x) = \sum_{n=1}^{\infty} b_n \frac{\sin n\pi x}{l}$$

$$b_n = \frac{2}{l} \int_0^l f(x) \frac{\sin n\pi x}{l} dx$$

$$= \frac{2}{l} \int_0^l x \frac{\sin n\pi x}{l} dx$$

$$u = x \quad dv = \frac{\sin n\pi x}{l}$$

$$u' = 1 \quad v = -\cos \frac{n\pi x}{l} \times \frac{l}{n\pi}$$

$$u'' = 0$$

$$v_1 = -\frac{\sin n\pi x}{l} \cdot \frac{l^2}{n^2\pi^2}$$

$$b_n = \frac{2}{l} \left[\left(x \cos \frac{n\pi x}{l} \cdot \frac{l}{n\pi} \right)_0^l + \left(\frac{\sin n\pi x}{l} \cdot \frac{l^2}{n^2\pi^2} \right)_0^l \right]$$

$$= \frac{2}{l} \left[\frac{-l}{n\pi} \left(+ x \cosh \frac{n\pi x}{l} \right)_0^l + \frac{l^2}{n^2 \pi^2} \left(\frac{\sin n\pi x}{l} \right)_0^l \right]$$

$$= \frac{2}{l} \left[\frac{-l}{n\pi} \left(+ l \cosh \frac{n\pi x}{l} - 0 \right) + \frac{l^2}{n^2 \pi^2} \left(\frac{\sin n\pi x}{l} - 0 \right) \right]$$

$$= \frac{2}{l} \left[\frac{-l}{n\pi} (l(-1)^n) + \frac{l^2}{n^2 \pi^2} (0) \right]$$

$$= \frac{2}{l} \left[\frac{-l}{n\pi} l(-1)^n \right]$$

$$b_n = \frac{-2l}{n\pi} l(-1)^n$$

Sub b_n value in $f(x)$

$$f(x) = \sum_{n=1}^{\infty} \frac{-2l}{n\pi} (-1)^n \frac{\sin n\pi x}{l} //$$

Half-Range Cosine Series [Even fn]

$(0, 2\pi)$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx,$$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$$

Half - Range Sine Series (odd f)

$(0, 2\pi)$

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx$$

$$l \Rightarrow nx = \frac{n\pi x}{l}$$

Problem:

1. Find the half Range cosine series of

$$f(x) = x \text{ in } (0, \pi)$$

w.k.T

Half Range cosine series is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx \rightarrow \textcircled{1}$$

To find a_0

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) \, dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x \, dx$$

$$= \frac{2}{\pi} \left(\frac{x^2}{2} \right)_0^{\pi}$$

$$= \frac{2}{\pi} \cdot \left(\frac{\pi^2}{2} \right)$$

$$\boxed{a_0 = \pi}$$

find a_n

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x \cos nx \, dx$$

$$dv = \cos nx$$

$$v = \frac{\sin nx}{n}$$

$$v_1 = -\frac{\cos nx}{n^2}$$

$$a_n = \frac{2}{\pi} \left[x \frac{\sin nx}{n} + \frac{\cos nx}{n^2} \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[\frac{1}{n} (x \sin nx)_0^{\pi} + \frac{1}{n^2} (\cos nx)_0^{\pi} \right]$$

$$= \frac{2}{\pi} \left\{ \frac{1}{n} (\pi \sin n\pi - 0) + \frac{1}{n^2} [\cos n\pi - \cos(0)] \right\}$$

$$= \frac{2}{\pi} \left[\frac{1}{n^2} [(-1)^n - 1] \right]$$

$$a_n = \frac{2}{\pi n^2} (-1)^n - 1$$

$$a_n = \begin{cases} 0, & n \text{ is even} \\ -\frac{4}{\pi n^2}, & n \text{ is odd} \end{cases}$$

$$a_n = \frac{-4}{\pi n^2}$$

Sub above values in equ (1)

$$f(x) = \frac{\pi}{2} + \sum_{n=1,3,5} \left(\frac{-4}{n^2\pi} \right) \cosh nx$$

$$f(x) = \frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1,3,5,\dots} \frac{\cosh nx}{n^2} //$$

3. Find the half range cosine series if

$$f(x) = e^x \text{ in } (0, \pi)$$

W.K.T

~~Power~~ series

Half range cosine series is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cosh nx \rightarrow (1)$$

To find a_0 :

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx$$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} e^x dx$$

$$= \frac{2}{\pi} (e^x)_0^{\pi}$$

$$= \frac{2}{\pi} (e^{\pi} - e^0)$$

$$a_0 = \frac{2(e^{\pi} - 1)}{\pi}$$

find a_n :

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} e^x \cos nx \, dx$$

v.k.T

$$\int e^{ax} \cos bx = \frac{e^{ax}}{a^2 + b^2} [a \cos bx + b \sin bx]$$

Here $a = 1$ $b = n$

$$a_n = \frac{2}{\pi} \left[\frac{e^x}{1+n^2} (\cos nx + n \sin nx) \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[\frac{e^{\pi}}{1+n^2} (\cos n\pi + n \sin n\pi) - \frac{e^0}{1+n^2} (\cos n(0) + n \sin n(0)) \right]$$

$$= \frac{2}{\pi} \left[\frac{e^{\pi}}{1+n^2} (-1)^n - \frac{1}{1+n^2} (1) \right]$$

$$= \frac{2}{\pi} \left(\frac{1}{1+n^2} [e^{\pi} (-1)^n - 1] \right)$$

$$a_n = \frac{2}{\pi(1+n^2)} [e^{\pi} (-1)^n - 1]$$

Sub a_0 & a_n in eqn (1)

$$f(x) = \frac{1}{2} \left[\frac{d}{dx} (e^{\pi x} - 1) \right] + \sum_{n=1}^{\infty} \frac{2}{\pi(1+n^2)} \frac{[e^{\pi(-1)^n} - 1]}{\cos nx}$$

$$f(x) = \frac{1}{\pi} (e^{\pi} - 1) + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{[e^{\pi(-1)^n} - 1]}{1+n^2} \cos nx //$$

(5)

4. Expand $f(x) = \cos x$ in $(0, \pi)$ by Fourier sine series.

A.T $f(x) = \cos x$

w.k.T

Fourier sine series in $(0, \pi)$ is

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx \rightarrow (1)$$

To find b_n

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx$$

$$= \frac{2}{\pi} \int_0^{\pi} \cos x \sin nx \, dx$$

$$= \frac{2}{\pi} \int_0^{\pi} \sin nx \cos x \, dx$$

w.k.T

$$\sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$$

$$\begin{aligned}
 &= \frac{2}{\pi} \int_0^{\pi} \left[\frac{1}{2} \sin(n\pi + x) + \sin(n\pi - x) \right] dx \\
 &= \frac{1}{\pi} \left[\int_0^{\pi} \sin(n+1)x dx + \int_0^{\pi} \sin(n-1)x dx \right] \\
 &= \frac{1}{\pi} \left\{ \left[-\frac{\cos(n+1)x}{n+1} \right]_0^{\pi} + \left[-\frac{\cos(n-1)x}{n-1} \right]_0^{\pi} \right\}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{\pi} \left\{ \frac{-1}{n+1} [\cos(n+1)\pi - \cos(n+1)0] \right. \\
 &\quad \left. - \frac{1}{n-1} [\cos(n-1)\pi - \cos(n-1)0] \right\}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{\pi} \left[\frac{-1}{n+1} (-(-1)^n - 1) - \frac{1}{n-1} (-(-1)^n - 1) \right] \\
 &= \frac{1}{\pi} \left[\frac{1}{n+1} [(-1)^n + 1] + \frac{1}{n-1} [(-1)^n + 1] \right]
 \end{aligned}$$

$$= \frac{1}{\pi} [(-1)^n + 1] \left(\frac{1}{n+1} + \frac{1}{n-1} \right)$$

$$= \frac{1}{\pi} [(-1)^n + 1] \frac{n-1 + n+1}{(n+1)(n-1)}$$

$$= \frac{1}{\pi} [(-1)^n + 1] \frac{2n}{n^2-1} \quad \sin A \cos B = \frac{1}{2}$$

$$b_n = \frac{2n}{\pi(n^2-1)} [(-1)^n + 1]$$

$$b_n = \begin{cases} \frac{4n}{\pi(n^2+1)} & n \text{ is even} \\ 0 & n \text{ is odd} \end{cases}$$

$$f(x) = \frac{4}{\pi} \sum_{n=2,4,6} \frac{n}{n^2+1} (\sin nx) //$$

2. Find the Half Range Sine Series if
 $f(x) = (x-1)^2$ in $(0, 1)$ (i.e.) $(0, l)$

W.K.T

Half Range Sine Series is

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}$$

Here $l=1$

$$f(x) = \sum_{n=1}^{\infty} b_n \sin n\pi x \rightarrow (1)$$

$$b_n = \frac{2}{l} \int_0^l f(x) \frac{\sin n\pi x}{1} dx$$

Here $l=1$

$$b_n = \frac{2}{1} \int_0^1 f(x) \sin n\pi x dx$$

$$= \frac{2}{1} \int_0^1 (x-1)^2 \sin n\pi x dx$$

$$u = (x-1)^2$$

$$dv = \sin n\pi x$$

$$u' = 2(x-1)(1)$$

$$V = \frac{-\cos n\pi x}{n\pi}$$

$$u'' = 2$$

$$V_1 = \frac{-\sin n\pi x}{n^2\pi^2}$$

$$V_2 = \frac{\cos n\pi x}{n^3\pi^3}$$

$$b_n = 2 \left[-(\alpha-1)^2 \frac{\cosh n\pi}{n\pi} + \alpha(\alpha-1) \frac{\sinh n\pi}{n^2\pi^2} + \frac{2 \cosh n\pi}{n^3\pi^3} \right]_0^1$$

$$b_n = 2 \left[\frac{-1}{n\pi} \left[(1-1)^2 \cosh n\pi - (0-1)^2 \cosh n\pi(0) \right] \right. \\ \left. + \frac{2}{n^2\pi^2} \left[(1-1) \sinh n\pi - (0-1) \sinh n\pi(0) \right] \right.$$

$$\left. + \frac{2}{n^3\pi^3} \left[\cosh n\pi - \cosh n\pi(0) \right] \right]$$

$$b_n = 2 \left[\frac{-1}{n\pi} (-1) + \frac{2}{n^3\pi^3} [(-1)^n - 1] \right] //$$

$$b_n = \begin{cases} 0, & n \text{ is even} \\ \frac{-4}{n^3\pi^3}, & n \text{ is odd} \end{cases}$$

$$b_n = 2 \left[\frac{1}{n\pi} - \frac{4}{n^3\pi^3} \right] \quad 2 \left[\frac{n^2\pi^2 - 4}{n^3\pi^3} \right]$$

Sub b_n in equ (1)

$$f(x) = 2 \sum_{n=1,3,5,\dots} \left[\frac{1}{n\pi} - \frac{4}{n^3\pi^3} \right] \sinh n\pi x //$$